Problem 4

Find the modulus and argument of the complex numbers:

$$1^\circ \; -i; \quad 2^\circ \; 3i; \quad 3^\circ \; 5; \quad 4^\circ \; 1+i; \quad 5^\circ \; 1-i; \quad 6^\circ \; 2+5i; \quad 7^\circ \; 3-4i;$$
$$8^\circ \; -3-5i.$$

Solution

For each complex number $z = x + iy$, we will find the modulus and argument.

Formulas

Modulus

The modulus (or absolute value) of $z = x + iy$ is:

$$|z| = \sqrt{x^2 + y^2}$$

Argument

The argument $\arg(z)$ is the angle θ such that:

$$x = |z| \cos\theta, \quad y = |z| \sin\theta$$

The argument can be found using:

$$\tan\theta = \frac{y}{x} \quad (\text{when } x \neq 0)$$

However, care must be taken to choose the correct quadrant:

- Quadrant I ($x > 0, y > 0$): $\theta = \arctan\left(\frac{y}{x}\right)$
- Quadrant II ($x < 0, y > 0$): $\theta = \pi - \arctan\left(\frac{|y|}{|x|}\right)$
- Quadrant III ($x < 0, y < 0$): $\theta = -\pi + \arctan\left(\frac{|y|}{|x|}\right)$ or $\pi + \arctan\left(\frac{y}{x}\right)$
- Quadrant IV ($x > 0, y < 0$): $\theta = -\arctan\left(\frac{|y|}{x}\right)$
- Positive real axis ($y = 0, x > 0$): $\theta = 0$
- Negative real axis ($y = 0, x < 0$): $\theta = \pi$
- Positive imaginary axis ($x = 0, y > 0$): $\theta = \frac{\pi}{2}$
- Negative imaginary axis ($x = 0, y < 0$): $\theta = -\frac{\pi}{2}$

Part 1°: $z = -i$

Standard Form

$z = 0 + i(-1)$, so $x = 0, y = -1$

Modulus

$$|z| = \sqrt{0^2 + (-1)^2} = \sqrt{1} = 1$$

Argument

The point $(0, -1)$ lies on the negative imaginary axis.

$$\arg(z) = -\frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

Answer

$$\boxed{|z| = 1, \quad \arg(z) = -\frac{\pi}{2} \text{ (or } -90)}$$

Part 2°: $z = 3i$

Standard Form

$z = 0 + i(3)$, so $x = 0, y = 3$

Modulus

$$|z| = \sqrt{0^2 + 3^2} = 3$$

Argument

The point $(0, 3)$ lies on the positive imaginary axis.

$$\arg(z) = \frac{\pi}{2}$$

Answer

$$\boxed{|z| = 3, \quad \arg(z) = \frac{\pi}{2} \text{ (or } 90)}$$

Part 3°: $z = 5$

Standard Form

$z = 5 + i(0)$, so $x = 5, y = 0$

Modulus

$$|z| = \sqrt{5^2 + 0^2} = 5$$

Argument

The point $(5, 0)$ lies on the positive real axis.

$$\arg(z) = 0$$

Answer

$$\boxed{|z| = 5, \quad \arg(z) = 0 \text{ (or } 0)}$$

Part 4°: $z = 1 + i$

Standard Form

$z = 1 + i(1)$, so $x = 1, y = 1$

Modulus

$$|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Argument

The point $(1, 1)$ is in Quadrant I with $\tan\theta = \frac{1}{1} = 1$.

$$\arg(z) = \arctan(1) = \frac{\pi}{4}$$

Answer

$$\boxed{|z| = \sqrt{2}, \quad \arg(z) = \frac{\pi}{4} \text{ (or 45)}}$$

Part $5°$: $z = 1 - i$

Standard Form

$z = 1 + i(-1)$, so $x = 1, y = -1$

Modulus

$$|z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Argument

The point $(1, -1)$ is in Quadrant IV with $\tan\theta = \frac{-1}{1} = -1$.

$$\arg(z) = -\arctan(1) = -\frac{\pi}{4}$$

Answer

$$\boxed{|z| = \sqrt{2}, \quad \arg(z) = -\frac{\pi}{4} \text{ (or } -45)}$$

Part $6°$: $z = 2 + 5i$

Standard Form

$z = 2 + i(5)$, so $x = 2, y = 5$

Modulus

$$|z| = \sqrt{2^2 + 5^2} = \sqrt{4 + 25} = \sqrt{29}$$

Argument

The point $(2, 5)$ is in Quadrant I.

$$\arg(z) = \arctan\left(\frac{5}{2}\right) \approx 1.190 \text{ rad} \approx 68.2$$

Answer

$$\boxed{|z| = \sqrt{29}, \quad \arg(z) = \arctan\left(\frac{5}{2}\right)}$$

Part 7°: $z = 3 - 4i$

Standard Form

$z = 3 + i(-4)$, so $x = 3, y = -4$

Modulus

$$|z| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Argument

The point $(3, -4)$ is in Quadrant IV.

$$\arg(z) = -\arctan\left(\frac{4}{3}\right) \approx -0.927 \text{ rad} \approx -53.1$$

Answer

$$\boxed{|z| = 5, \quad \arg(z) = -\arctan\left(\frac{4}{3}\right)}$$

Part 8°: $z = -3 - 5i$

Standard Form

$z = -3 + i(-5)$, so $x = -3, y = -5$

Modulus

$$|z| = \sqrt{(-3)^2 + (-5)^2} = \sqrt{9 + 25} = \sqrt{34}$$

Argument

The point $(-3, -5)$ is in Quadrant III.

$$\arg(z) = -\pi + \arctan\left(\frac{5}{3}\right) \approx -\pi + 1.030 \approx -2.112 \text{ rad} \approx -121.0$$

Alternatively: $\arg(z) = \pi + \arctan\left(\frac{-5}{-3}\right) = \pi + \arctan\left(\frac{5}{3}\right)$
Or more simply: $\arg(z) \approx -2.112$ rad or equivalently ≈ 4.171 rad (if using principal value in $[0, 2\pi)$)

Answer

$$\boxed{|z| = \sqrt{34}, \quad \arg(z) = -\pi + \arctan\left(\frac{5}{3}\right)}$$

Summary Table

No.	Complex Number	Modulus	Argument
1°	$-i$	1	$-\frac{\pi}{2}$
2°	$3i$	3	$\frac{\pi}{2}$
3°	5	5	0
4°	$1 + i$	$\sqrt{2}$	$\frac{\pi}{4}$
5°	$1 - i$	$\sqrt{2}$	$-\frac{\pi}{4}$
6°	$2 + 5i$	$\sqrt{29}$	$\arctan(\frac{5}{2})$
7°	$3 - 4i$	5	$-\arctan(\frac{4}{3})$
8°	$-3 - 5i$	$\sqrt{34}$	$-\pi + \arctan(\frac{5}{3})$

Final Answer

$$\boxed{\begin{aligned}
1^\circ\ \ -i:&\quad |z|=1,\quad \arg(z)=-\frac{\pi}{2}\\
2^\circ\ 3i:&\quad |z|=3,\quad \arg(z)=\frac{\pi}{2}\\
3^\circ\ 5:&\quad |z|=5,\quad \arg(z)=0\\
4^\circ\ 1+i:&\quad |z|=\sqrt{2},\quad \arg(z)=\frac{\pi}{4}\\
5^\circ\ 1-i:&\quad |z|=\sqrt{2},\quad \arg(z)=-\frac{\pi}{4}\\
6^\circ\ 2+5i:&\quad |z|=\sqrt{29},\quad \arg(z)=\arctan\left(\frac{5}{2}\right)\\
7^\circ\ 3-4i:&\quad |z|=5,\quad \arg(z)=-\arctan\left(\frac{4}{3}\right)\\
8^\circ\ \ -3-5i:&\quad |z|=\sqrt{34},\quad \arg(z)=-\pi+\arctan\left(\frac{5}{3}\right)
\end{aligned}}$$